

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			{Peaks/maximum values} on absorption spectrum are at similar <u>wavelengths</u> to peaks on action spectrum/ At <u>wavelengths</u> (of light) where absorption {is higher/ increases} the rate of photosynthesis {is higher/ increases}/ora/ <u>owtte</u> (1) - Wavelengths (most) absorbed are the wavelengths used in photosynthesis. (1) Allow correct reference to values		1	1	2		2
	(b)	(i)		Any one (x1) from - Chlorophyll b has a peak {at approx. 640/ in the red (wavelengths)} but chlorophyll c does not (1) - Chlorophyll b absorbs longer wavelengths of light (1) - Chlorophyll b absorption increases around approx. 600 but chlorophyll c does not (1) - Chlorophyll c has one peak whereas Chlorophyll b has two (1) - Chlorophyll c absorbs less {at 640nm/ in the red (wavelengths)}/ ORA (1)		1		1		1
		(ii)		Any one (x1) from {No peak/ lower rate of photosynthesis} at 640nm (1) - Narrower peak {at 670nm/ in red region} (1)			1	1		1
	(c)			Any one (x1) from {Orange light/light of wavelength approx. 640} {does not penetrate water column/ does not reach 200m (use of figures)} (1) (Advantage to have chlorophyll c) as blue light penetrates {further / to 300m / 200m} (use of figures) (1)			1	1		
	(d)			Domain-Eukaryota/e (1) Reason: has {nucleus/chloroplast/membranous/ membrane bound organelles} (1)		2		2		